

# BioSpecInfo

## Traceability Matrix — BioSpecInfo

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# Traceability Matrix — BioSpecInfo

| Field             | Value                                                                                                        |
|-------------------|--------------------------------------------------------------------------------------------------------------|
| Software          | BioSpecInfo                                                                                                  |
| Author            | Samuele Pio Provenzano                                                                                       |
| Version described | bsi-v185                                                                                                     |
| Purpose           | Connect every declared requirement to the implementation that realises it and to the bench that verifies it. |

**How to read this matrix.** Every row is a closed chain: a requirement, the point in the code that realises it, and the bench that verifies it works. A row without a bench is an **unverified** requirement and is declared as such, not hidden: the usefulness of a matrix lies precisely in making the gaps visible.

The outcomes of the benches cited here are reported in full, with their complete output, in [../evidence/RAPPORTO-VERIFICA.md](#), which is generated automatically.

## 1. Scientific requirements

| ID     | Requirement                                                                                  | Implementation                                                                                                        | Verification bench                                                                         |
|--------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| SCI-01 | Structural data for drugs must be consistent with the declared molecular weight              | FARM_DATA in <code>index.html</code>                                                                                  | <code>tools/verifica-farmaci.js</code> — RDKit<br>↔ literature comparison over 233 entries |
| SCI-02 | Functional groups must be recognised on the structure, not on the text of the SMILES         | <code>bsi-spettri.js</code> §1, SMARTS patterns via RDKit                                                             | <code>test_spettri</code> — 8 reference molecules, additive recognition                    |
| SCI-03 | A molecule with several functional groups must show them all                                 | <code>bandeDaGruppi()</code> , additive construction                                                                  | <code>test_spettri</code> §1 — acetylsalicylic acid: ester <b>and</b> acid                 |
| SCI-04 | Spectra must respect the conventions of their technique                                      | <code>svgIR()</code> , <code>makeNMRsvg()</code> , <code>drawUVSpectrum()</code> , <code>drawMSSpectrum()</code>      | <code>test_assi</code> (SVG), <code>test_assi_canvas</code> (canvas)                       |
| SCI-05 | A computed spectrum must be reproducible                                                     | every <code>Math.random()</code> removed from the traces                                                              | <code>test_assi</code> §3, <code>test_spettri</code> §8 — byte-for-byte comparison         |
| SCI-06 | The height of <sup>1</sup> H peaks must represent the integration, or not be asserted at all | <code>makeNMRsvg()</code> ; <code>_nmrSvg()</code> draws bands of uniform height                                      | <code>test_assi</code> §4                                                                  |
| SCI-07 | Conjugation must act on the individual carbonyl                                              | SMARTS <code>acidoConiug</code> , <code>estereConiug</code> , <code>chetoneConiug</code> , <code>aldeideConiug</code> | <code>test_spettri</code> §1 — aspirin 1760/1690; acetophenone 1690                        |

| ID     | Requirement                                                                                                | Implementation                                                                                                  | Verification bench                                                                                                                                                                                |
|--------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI-08 | An ambiguous physical constant must not be resolved arbitrarily                                            | <code>costante_fisica</code> in <code>bsi-ai-hub.js</code> — collects every candidate                           | <code>test_costanti</code> — 45 checks, including the refusals for ambiguity                                                                                                                      |
| SCI-09 | The limits of the predictors must be declared to the user                                                  | chart headers; <code>docs/06 §3.5</code>                                                                        | <code>verifica_guida</code> — consistency between promises and code                                                                                                                               |
| SCI-10 | Molecular symmetry must determine the selection rules correctly                                            | <code>ssimm</code> section in <code>index.html</code>                                                           | <code>test_simmetria</code> — 26 checks: normal modes, centre of inversion, mutual exclusion                                                                                                      |
| SCI-12 | A chromatogram must follow from the parameters, not be drawn by hand                                       | <code>scroma</code> section: $\sigma = t \sqrt{R}$ , $R$ from distance and widths                               | <code>audit_stabilita</code> (opening and drawing), inspection of the values against the definitions                                                                                              |
| SCI-13 | A calibration line must declare the uncertainty of its coefficients and recognise the two ways it misleads | <code>staratura</code> section: ordinary least squares, F-test line $\rightarrow$ parabola, deleted residual    | Verified on the three example sets included: linear ( $F = 0.3$ ), non-linear ( $F = 186.6 > 7.71$ ), with an outlier (1 detected)                                                                |
| SCI-14 | A QSAR model must be evaluated on scaffolds never seen, not on analogues from the same set                 | <code>bsi-cheminfo.js §9</code> — <code>divisionePerScaffold()</code> alongside <code>divisioneCasuale()</code> | <code>test_cheminfo</code> — the scaffold split leaves no scaffold shared between training and test                                                                                               |
| SCI-15 | A model score must be compared against chance, not presented on its own                                    | <code>bsi-cheminfo.js §9</code> — retraining on scrambled labels, eight repetitions                             | <code>test_cheminfo</code> — verified in <b>both directions</b> : on a learnable signal $R^2$ 0.394 beats all eight twins (margin 0.287); on random labels $R^2$ -0.199 does <b>not</b> beat them |
| SCI-16 | Two spellings of the same molecule must not be counted twice, nor land on opposite sides of the split      | <code>bsi-cheminfo.js §1</code> — deduplication on the canonical SMILES after the largest fragment              | <code>test_cheminfo</code> — <code>0C(=O)C</code> and <code>CC(=O)O</code> collapse into one entry; salts lose the counterion                                                                     |
| SCI-17 | Similarity between molecules must be computed, not estimated                                               | <code>bsi-cheminfo.js §3</code> — Tanimoto and Dice over Morgan, RDKit, pattern and MACCS fingerprints          | <code>test_cheminfo</code> — compared against hand-computable values, including the $0/0 = 1$ convention                                                                                          |
| SCI-18 | A validation on a single split, over a small set, is a noisy number: it must come with its dispersion      | <code>bsi-cheminfo.js §9-bis</code> — <code>validazioneIncrociata()</code> over <b>scaffold-grouped</b> folds   | <code>test_cheminfo</code> — the folds cover every molecule once, <b>no scaffold sits in two folds</b> , and the same seed gives the same folds                                                   |
| SCI-19 | A result that changes on every run is verifiable by nobody                                                 | seeded mulberry32 generator for splits, scrambles and folds; the seed is written into the report                | <code>test_cheminfo</code> — two calls with the same seed give identical folds; the method report declares the seed                                                                               |

| ID     | Requirement                                                                                                                                                                                      | Implementation                                                                                                            | Verification bench                                                                                                                                                                                                                                                                    |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI-20 | Results must be able to leave the tool, and the method must leave with them                                                                                                                      | bsi-cheminfo.js §9-ter — esportaCsv() and rapportoMetodo(), both formed in memory in the browser                          | test_cheminfo — columns consistent with the header, a name containing a comma quoted, residual computed, empty cell for missing activity; the report carries seed, fingerprint, verdict, limits                                                                                       |
| SCI-21 | A number presented as structural similarity must come from a fingerprint, and one implementation must compute it                                                                                 | rdkit_lab.html loads bsi-cheminfo.js and uses its tanimoto() and fingerprint(); the eight-bit similarity has been removed | test_cheminfo — page and engine give the same Tanimoto over all <b>28 pairs</b> , deviation <b>exactly 0</b> ; the fake-similarity functions no longer exist; every reference structure is valid and carries the <b>monoisotopic mass</b> of the drug it names                        |
| SCI-22 | Conversions between molecular languages must be checked against an external source, the IUPAC namer must refuse what it cannot name, and coordinates written to a file must be real and declared | bsi-molingue.js — <b>eight languages</b> plus XYZ, PDB, CXSMARTS and CIP descriptors                                      | test_lingue — <b>7 literature InChI keys written inside the bench</b> , full SMILES → molfile → SMILES round-trip, 26 hand-written names in two languages, <b>7 required refusals</b> ; XYZ and PDB with non-null coordinates <b>and</b> declared 2D; a reaction recognised and split |
| SCI-23 | Angles, lengths and dihedrals are measured on the coordinates, and on flat coordinates they are not shown                                                                                        | dot product for angles, cross product for dihedrals                                                                       | test_mol3d — tetrahedron <b>109.4712°</b> , water 104.47°, CO <sub>2</sub> 180°, BF <sub>3</sub> 120°, ammonia 106.13°; dihedrals 0°, 60°, 90°, 180°; ethanol C-C 1.509 Å; and the refusal tested in both directions                                                                  |

## 2. AI agent requirements

| ID    | Requirement                                                                 | Implementation                                        | Verification bench            |
|-------|-----------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------|
| AI-01 | A model retired by the provider must not block the assistant                | modelloSuggestitoDaErrore(), bsi_modelli_ko blacklist | test_404 — 36 checks          |
| AI-02 | A provider unreachable from the browser must be remembered and demoted      | bsi_prov_ko, 24 h TTL, providerUtilizzabili()         | test_ko (35), browser_ko (37) |
| AI-03 | A temporary overload must produce a retry, not an error                     | erroreTemporaneo(), TENTATIVI_TEMPORANEO              | test_503 — 40 checks          |
| AI-04 | An imposed wait longer than 10 s must switch provider if a ready one exists | ATTESA_TROPPO_LUNGA_MS                                | test_attesalunga — 14 checks  |
| AI-05 | The provider's token ceiling must be learned from the error, not hard-coded | tettoDaErrore(), MARGINE_TETTO                        | test_tetto — 32 checks        |

| ID    | Requirement                                                              | Implementation                                       | Verification bench                                                                                          |
|-------|--------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| AI-06 | Opaque reasoning signatures must be returned to the provider             | capture and replay in <code>appendAgentTurn()</code> | <code>test_firma</code> — 31 checks                                                                         |
| AI-07 | Two overlapping submissions must not corrupt the history                 | <code>_turnoInCorso</code> in <code>send()</code>    | <code>test_doppioinvio</code> — 7 checks                                                                    |
| AI-08 | The user must be able to measure which providers they can actually reach | <code>provaTuttiIFornitori()</code>                  | <code>browser_prova</code> — 29 checks                                                                      |
| AI-09 | Paid providers must not be offered as free ones                          | separation by <code>PROVIDERS[id].free</code>        | <code>browser_prova</code> §D2                                                                              |
| AI-10 | A documented route must exist to remove CORS and rate limits             | <code>proxy/spectra-proxy.js</code> + in-app guide   | <code>test_guidaproxy</code> (12),<br><code>browser_proxy</code> (15),<br><code>browser_proxyui</code> (17) |

### 3. Stability requirements

| ID     | Requirement                                                         | Implementation                                                | Verification bench                                                        |
|--------|---------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------|
| STA-01 | A long session must accumulate neither DOM nodes nor timers         | section lifecycle management                                  | <code>audit_stabilita</code> §1 — 91 sections × 5 rounds                  |
| STA-02 | Exhausting <code>localStorage</code> must not make the app unusable | guarded writes; persistent warning in the File Manager        | <code>audit_quota</code> (10),<br><code>test_filemanager</code> (15)      |
| STA-03 | No rejected promise may remain unhandled                            | systematic <code>.catch()</code>                              | <code>audit_promesse</code> — 22 checks, healthy network and dead network |
| STA-04 | Corrupted saved data must not prevent startup                       | <code>loadJSON()</code> with fallback                         | <code>audit_stabilita</code> §4                                           |
| STA-05 | A slow network must not be treated as an absent one                 | <code>ATTESA_RETE_MS = 3500</code> in <code>sw.js</code>      | <code>test_sw</code> §3 and §3-bis — measured 3,507 ms                    |
| STA-06 | An update must not destroy work in progress                         | <code>BSI_SW_UPDATED</code> message, decision taken page-side | <code>test_sw</code> §4                                                   |
| STA-07 | Browsing molecules must not exhaust the WebGL contexts              | reuse of the 3D viewer at all 5 creation points               | <code>test_visore3d</code> — from 7 constructions down to 1               |
| STA-08 | An unavailable IndexedDB store must be declared, not fatal          | <code>dbStore()</code> over a promise; persistent warning     | <code>test_filemanager</code> §2                                          |

## 4. Interface and accessibility requirements

| ID    | Requirement                                                                                                                                | Implementation                                                                                                                         | Verification bench                                                                                                                                                                                                                                     |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UI-01 | Every page must open without JavaScript errors                                                                                             | —                                                                                                                                      | audit_stabilita \$5 — 14 pages                                                                                                                                                                                                                         |
| UI-02 | Charts must be sharp on high-density screens                                                                                               | bsiNitido() on the canvas contexts                                                                                                     | audit_grafici — 40 canvases                                                                                                                                                                                                                            |
| UI-03 | The app must work at 390 px width                                                                                                          | auto-fit grids, no fixed column                                                                                                        | audit_mobile — the document's scrollWidth across all 91 sections at 390 px; audit_stabilita for errors in a phone viewport                                                                                                                             |
| UI-04 | The user must be able to know which version they are running and force the update                                                          | "Updates" entry in the ✨ panel                                                                                                         | test_aggiorna — 9 checks                                                                                                                                                                                                                               |
| UI-05 | Data deletion must be selective and reversible in its choices                                                                              | bsiCancellaDati() by group                                                                                                             | browser_reset — 24 checks                                                                                                                                                                                                                              |
| UI-08 | Surfaces that appear only after an action must also meet WCAG AA contrast                                                                  | the six panels of the Cheminformatics section                                                                                          | test_cheminfo — the analysis is run, the panels opened in turn, <b>460 text elements measured, 0 defects</b> , with the WCAG formulas rewritten inside the bench                                                                                       |
| UI-09 | Opening a section must not block the page                                                                                                  | 3Dmol's first render() moved out of the click; the 296 synthesis figures drawn in 8 ms slices                                          | test_fluidita — <b>91 sections traversed</b> , median switch 14 ms, worst 162 ms, <b>0 sections above 100 ms</b> ; plus the opposite proofs: the WebGL canvas appears anyway, all 296 figures exist shortly after, and printing does not come out mute |
| UI-10 | The interface must switch among many languages, declaring how much each is translated, and the feature must be reachable from every device | bsi-lingue.js with <b>twelve languages</b> and a searchable selector; two navigation sections, because on a phone the ✨ menu is hidden | test_lingue — <b>162 skeleton elements translated out of 162 in each of the twelve languages</b> , coverage computed by applying the dictionary and counting; the text really changes and comes back identical; search tested in both directions       |
| UI-11 | The molecule must show how it holds together, and the viewer must choose the angle                                                         | bsi-mol3d.js: formation from dust with staggered arrival, and atom selection by click                                                  | test_mol3d — the canvas <b>pixels</b> at 130 ms and 2.3 s, the arrival order (skeleton before hydrogens) measured on each atom's progress, and the selection of 1, 2, 3 and 4 atoms                                                                    |

## 5. Security and privacy requirements

| ID     | Requirement                                                                          | Implementation                                                                                    | Verification bench                                                                                                                                                                                                                                                                            |
|--------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEC-01 | No API key may be present in the repository                                          | keys only in <code>localStorage</code> or in the Worker's secrets                                 | <code>tools/verifica-sicurezza.js</code> — 295 tracked files, 8 credential shapes                                                                                                                                                                                                             |
| SEC-02 | No personal data may leave the device without an explicit action                     | local-first architecture, telemetry disabled                                                      | <code>audit_rete</code> — <b>direct verification</b> : a canary value seeded into 71 stores of the user's data, the application used across 91 sections on 6 pages, and the URL, headers and body of every request inspected. Plus <code>verifica-sicurezza</code> on the two exit mechanisms |
| SEC-03 | Passwords must not appear in clear text in the source                                | SHA-256 in <code>file_manager.html</code>                                                         | <code>tools/verifica-sicurezza.js</code>                                                                                                                                                                                                                                                      |
| SEC-07 | No credential may EVER have entered the repository, not even in a commit later fixed | no key has ever been committed; keys live in <code>localStorage</code> or in the Worker's secrets | <code>audit_storia</code> — <b>1,551 distinct versions of text files across 390 commits</b> , 9 patterns tested in both directions; fails on a shallow clone, because it would measure less surface                                                                                           |
| SEC-04 | No conflict marker may reach publication                                             | —                                                                                                 | <code>verifica_guida</code> §12 — 58 text files                                                                                                                                                                                                                                               |

**SEC-03 — declared limitation.** GitHub Pages serves static files: any page-side access control can be bypassed by whoever reads the source. The File Manager's protection is a deterrent, **not** a security control, and the documentation says so. Furthermore the password remained in clear text in the git history until its removal, and taking a secret out of the files does not take it out of the history: `git log -p` hands it to anyone. The only effective remedy was to change it, and that **has been done** at version `bsi-v185`. The old one remains in the history and no longer opens anything.

## 6. Requirements not yet verified automatically

Declared for completeness. They are covered by documentary inspection or by manual testing, and their automation is planned.

| ID    | Requirement                  | Current coverage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UI-06 | Full WCAG 2.1 AA conformance | the mechanical part is automated across <b>13 pages and 91 sections</b> ( <code>tools/verifica-accessibilita.js</code> , 33,642 text elements: it was 19,751 before text over gradients entered the measurement). <b>Of the 1,069 contrast defects that emerged, none remain: 0 measured</b> over those same 91 sections, and the value is recorded as a baseline the bench defends. Unlabelled fields are <b>zero</b> : see the box in <code>docs/09</code> §4. What remains outside is text inside SVGs and text over a real background image, counted on every run (D-09) |

| ID             | Requirement                                                      | Current coverage                                                                                                                                                                                    |
|----------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PERF-01</b> | First-paint time on a low-end device                             | manual cross-device testing ( docs/02 §4)                                                                                                                                                           |
| <b>SCI-11</b>  | Structures of 2 entries that are not single molecules (it was 6) | <b>not representable</b> : Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. The other four were closed by taking the structure from ChEMBL, see docs/06 §2.4 |
| <b>PERF-02</b> | Code coverage of the verification benches                        | <b>partially measured</b> : 49.89 % of statements over the widest path ( audit_copertura ). Whole-battery coverage and branch coverage remain unmeasured; see §8                                    |
| <b>UI-07</b>   | Behaviour on Firefox and WebKit                                  | <b>not verified</b> — the battery runs on Chromium only; see §8                                                                                                                                     |

## 7. Overall coverage

| Category                 | Requirements | Verified by a bench | Coverage     |
|--------------------------|--------------|---------------------|--------------|
| Scientific (SCI-01...23) | 22           | 22                  | 100 %        |
| AI agent (AI-01...10)    | 10           | 10                  | 100 %        |
| Stability (STA-01...08)  | 8            | 8                   | 100 %        |
| Interface (UI-01...11)   | 9            | 9                   | 100 %        |
| Security (SEC-01...07)   | 5            | 5                   | 100 %        |
| <b>Automated total</b>   | <b>54</b>    | <b>54</b>           | <b>100 %</b> |
| Not automated (§6)       | 5            | 0                   | 0 %          |
| <b>Declared total</b>    | <b>59</b>    | <b>54</b>           | <b>91 %</b>  |

Coverage is computed over the requirements **declared in this document** and must not be confused with code coverage: it measures how many requirements have a bench verifying them, not how many lines are executed during the tests.

**Why two totals.** Reporting only the 100 % of automated requirements would be true and misleading at once: it is 100 % of what was *chosen* to be automated. The number that matters to an assessor is the second one — **54 requirements verified out of 59 declared** — and the five that are missing are listed by name in §6, not summarised into a percentage.

This table is checked by `tools/verifica-documenti.js`, which counts the identifiers actually present in the document and fails if the totals written here do not match. It is a check born of a real error: the totals said 36 over 9 scientific categories while the identifiers defined were 10, and `SCI-10` appeared twice — once as a verified requirement and once as an unverified one.

## 8. Declared coverage gaps

These are not defects: they are things that have **not been measured**, and that therefore cannot be asserted.

| Gap                                  | What it entails                                                                                                                                            | Why it is so                                                                                                                                                                                                                            |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Partial code coverage</b>         | Measured: <b>49.89 %</b> of statements over the widest path a bench walks. It is not the coverage of the whole battery, and it is statements, not branches | Chromium's profiler collects it inside the engine, with no build and without rewriting the source: the obstacle belonged to the tool (c8, istanbul), not to the problem. Value recorded; <code>audit_copertura</code> fails if it falls |
| <b>Chromium only</b>                 | Behaviour on Firefox and WebKit is verified by hand, not by a bench                                                                                        | The battery uses <code>playwright-core</code> , which downloads a single engine. In the verification environment the CDN for the other engines answers <b>403</b> to the network policy: they cannot be installed there                 |
| <b>No visual regression</b>          | An unintended graphical change would not be caught                                                                                                         | Spectra are deterministic and comparable byte for byte (SCI-05): the comparison exists on the traces, not on the whole page                                                                                                             |
| <b>2 entries without a structure</b> | Two entries out of 178 have no structure to show: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. It was six       | See <code>06-Scientific-Accuracy-Data-Provenance.md</code> §2.4                                                                                                                                                                         |
| <b>Performance on slow devices</b>   | First-paint time is not measured on low-end hardware                                                                                                       | It requires physical devices; the test is manual                                                                                                                                                                                        |

\_Document updated to version `bsi-v185`.\_